

layout.tsx.txt

```
import type { Metadata } from 'next';
import './globals.css';

export const metadata: Metadata = {
  title: 'AI-Enabled Skill Intelligence & Personalized Learning Platform | MoSPI NSSTA',
  description: 'Government enterprise competency intelligence, transparent skill-gap scoring, grounded RAG recommendations, and career navigator.',
};

export default function RootLayout({
  children,
}): {
  children: React.ReactNode;
}) {
  return (
    <html lang="en">
      <body>{children}</body>
    </html>
  );
}
```

globals.css

```
@tailwind base;
@tailwind components;
@tailwind utilities;

@layer base {
  body {
    @apply bg-slate-50 text-slate-900 antialiased font-sans;
  }
}

.glass-card {
  background: rgba(255, 255, 255, 0.9);
  backdrop-filter: blur(12px);
}
```

```
border: 1px solid rgba(226, 232, 240, 0.8);
}
```

Navbar.tsx.txt

```
"use client";
```

```
import React from 'react';
```

```
import { ShieldCheck, UserCheck, Bell, ChevronDown, Award } from 'lucide-react';
```

```
interface NavbarProps {
```

```
  user: any;
```

```
  currentRoleMode: 'learner' | 'admin';
```

```
  onToggleRoleMode: () => void;
```

```
}
```

```
export default function Navbar({ user, currentRoleMode, onToggleRoleMode }: NavbarPr  
ops) {
```

```
  return (
```

```
    <header className="bg-slate-900 border-b border-slate-800 text-white sticky top-0 z-  
40 px-6 py-3.5 flex items-center justify-between shadow-md">
```

```
      { /* Brand & Logo */ }
```

```
      <div className="flex items-center space-x-4">
```

```
        <div className="flex items-center space-x-2.5">
```

```
          <div className="bg-gradient-to-tr from-blue-600 to-indigo-500 p-2 rounded-xl text-  
white shadow-lg">
```

```
            <ShieldCheck className="w-6 h-6" />
```

```
          </div>
```

```
          <div>
```

```
            <div className="flex items-center space-x-2">
```

```
              <span className="font-extrabold text-lg tracking-tight text-white">SkillIntelligenc  
e</span>
```

```
              <span className="text-[10px] uppercase font-bold bg-blue-500/20 text-blue-300  
border border-blue-400/30 px-2 py-0.5 rounded-full">
```

```
                MoSPI / NSSTA
```

```
              </span>
```

```
            </div>
```

```
            <p className="text-[11px] text-slate-400 font-medium">iGOT Karmayogi Compete  
ncy Engine</p>
```

</div>

</div>

</div>

{/\* User Info & Role Switcher \*/}

<div className="flex items-center space-x-4">

{/\* Role Switcher Pill \*/}

<button

onClick={onToggleRoleMode}

className={`flex items-center space-x-2 px-3.5 py-1.5 rounded-lg text-xs font-semi  
bold border transition \${  
currentRoleMode === 'admin'  
? 'bg-amber-500/20 text-amber-300 border-amber-500/40 hover:bg-amber-500/30'  
0'

: 'bg-blue-500/20 text-blue-300 border-blue-500/40 hover:bg-blue-500/30'

}`}

>

<UserCheck className="w-4 h-4" />

<span>{currentRoleMode === 'admin' ? 'Switch to Learner View' : 'Switch Role (Admin)'}</span>

</button>

{/\* Notifications \*/}

<button className="relative p-2 text-slate-300 hover:text-white hover:bg-slate-800 rounded-lg transition">

<Bell className="w-5 h-5" />

<span className="absolute top-1.5 right-1.5 w-2 h-2 bg-blue-500 rounded-full animate-ping"></span>

<span className="absolute top-1.5 right-1.5 w-2 h-2 bg-blue-500 rounded-full"></span>

</button>

{/\* Profile Dropdown Badge \*/}

<div className="flex items-center space-x-3 pl-3 border-l border-slate-800">

<div className="w-9 h-9 rounded-full bg-gradient-to-tr from-blue-700 to-indigo-600 flex items-center justify-center font-bold text-sm text-white shadow-inner border border-blue-400/30">

{user?.name ? user.name.split(' ').map((n: string) => n[0]).join('') : 'AS'}

```

    </div>
    <div className="text-left hidden md:block">
      <span className="text-xs font-bold text-white block">{user?.name || 'Ananya Shar
ma'}</span>
      <span className="text-[11px] text-slate-400 block">{user?.designation || 'Statistic
al Officer'}</span>
    </div>
  </div>
</div>
</header>
);
}

```

compentancyRadarChart.tsx.txt

"use client";

```

import React from 'react';
import {
  Radar, RadarChart, PolarGrid, PolarAngleAxis, PolarRadiusAxis, ResponsiveContainer, ToolTip
} from 'recharts';

```

```

interface CompetencyRadarChartProps {
  data?: any[];
}

```

```

const defaultData = [
  { domain: 'Statistical', assessed: 4, required: 4, fullMark: 5 },
  { domain: 'Technical', assessed: 1, required: 3, fullMark: 5 },
  { domain: 'Digital Governance', assessed: 3, required: 3, fullMark: 5 },
  { domain: 'Behavioural', assessed: 2, required: 2, fullMark: 5 },
  { domain: 'Data', assessed: 2, required: 3, fullMark: 5 },
  { domain: 'Leadership', assessed: 1, required: 2, fullMark: 5 }
];

```

```

export default function CompetencyRadarChart({ data }: CompetencyRadarChartProps) {
  const chartData = data && data.length > 0 ? data : defaultData;

```

```

return (
  <div className="w-full h-[280px]">
    <ResponsiveContainer width="100%" height="100%">
      <RadarChart cx="50%" cy="50%" outerRadius="75%" data={chartData}>
        <PolarGrid stroke="#e2e8f0" />
        <PolarAngleAxis dataKey="domain" tick={{ fill: '#475569', fontSize: 11, fontWeight: 600 }} />
        <PolarRadiusAxis angle={30} domain={[0, 5]} tick={{ fill: '#94a3b8', fontSize: 10 }} />

        <Radar
          name="Assessed Capability"
          dataKey="assessed"
          stroke="#2563eb"
          fill="#3b82f6"
          fillOpacity={0.4}
        />
        <Radar
          name="Required Level"
          dataKey="required"
          stroke="#dc2626"
          fill="#ef4444"
          fillOpacity={0.15}
          strokeDasharray="4 4"
        />
        <Tooltip />
      </RadarChart>
    </ResponsiveContainer>

    <div className="flex items-center justify-center space-x-6 text-xs text-slate-600 mt-1 font-medium">
      <div className="flex items-center space-x-2">
        <span className="w-3 h-3 rounded-full bg-blue-500 inline-block"></span>
        <span>Assessed Level</span>
      </div>
      <div className="flex items-center space-x-2">
        <span className="w-3 h-3 rounded-full border border-red-500 bg-red-100 inline-block"></span>

```

```
        <span>Role Required Benchmark</span>
    </div>
</div>
</div>
);
}
AdminHeatmap.tsx.txt

"use client";

import React from 'react';

interface HeatmapProps {
  data: any[];
}

export default function AdminHeatmap({ data }: HeatmapProps) {
  const getSeverityBg = (gapAvg: number) => {
    if (gapAvg <= 0.3) return 'bg-emerald-500 text-white';
    if (gapAvg <= 0.8) return 'bg-yellow-400 text-slate-900';
    if (gapAvg <= 1.5) return 'bg-orange-500 text-white';
    return 'bg-red-600 text-white font-bold';
  };

  return (
    <div className="overflow-x-auto">
      <table className="w-full text-xs text-left border-collapse">
        <thead>
          <tr className="bg-slate-900 text-white">
            <th className="p-3 font-semibold rounded-tl-lg">Government Role</th>
            <th className="p-3 font-semibold">Department</th>
            <th className="p-3 text-center font-semibold">Statistical</th>
            <th className="p-3 text-center font-semibold">Technical</th>
            <th className="p-3 text-center font-semibold">Digital Gov</th>
            <th className="p-3 text-center font-semibold">Behavioural</th>
            <th className="p-3 text-center font-semibold">Data</th>
            <th className="p-3 text-center font-semibold rounded-tr-lg">Leadership</th>
          </tr>
```

```

</thead>
<tbody className="divide-y divide-slate-200">
  {data && data.length > 0 ? (
    data.map((item, idx) => (
      <tr key={idx} className="hover:bg-slate-50 transition">
        <td className="p-3 font-bold text-slate-900">{item.role_name}</td>
        <td className="p-3 text-slate-500">{item.department}</td>
        <td className="p-2 text-center">
          <span className={`px-2.5 py-1 rounded-md text-xs inline-block w-12 text-center`} ${{getSeverityBg(item.statistical_avg_gap)}}`>
            {item.statistical_avg_gap.toFixed(1)}
          </span>
        </td>
        <td className="p-2 text-center">
          <span className={`px-2.5 py-1 rounded-md text-xs inline-block w-12 text-center`} ${{getSeverityBg(item.technical_avg_gap)}}`>
            {item.technical_avg_gap.toFixed(1)}
          </span>
        </td>
        <td className="p-2 text-center">
          <span className={`px-2.5 py-1 rounded-md text-xs inline-block w-12 text-center`} ${{getSeverityBg(item.governance_avg_gap)}}`>
            {item.governance_avg_gap.toFixed(1)}
          </span>
        </td>
        <td className="p-2 text-center">
          <span className={`px-2.5 py-1 rounded-md text-xs inline-block w-12 text-center`} ${{getSeverityBg(item.behavioural_avg_gap)}}`>
            {item.behavioural_avg_gap.toFixed(1)}
          </span>
        </td>
        <td className="p-2 text-center">
          <span className={`px-2.5 py-1 rounded-md text-xs inline-block w-12 text-center`} ${{getSeverityBg(item.data_avg_gap)}}`>
            {item.data_avg_gap.toFixed(1)}
          </span>
        </td>
        <td className="p-2 text-center">

```

```

        <span className={`px-2.5 py-1 rounded-md text-xs inline-block w-12 text-center`}
        r ${getSeverityBg(item.leadership_avg_gap)}`}>
            {item.leadership_avg_gap.toFixed(1)}
        </span>
    </td>
</tr>
))
):(
    <tr>
        <td colspan={8} className="p-4 text-center text-slate-400">Loading Heatmap...<
/td>
    </td>
    </tr>
    </tbody>
</table>

```

```

<div className="flex items-center justify-end space-x-4 text-[11px] text-slate-500 mt-3 font-medium">
    <div className="flex items-center space-x-1.5">
        <span className="w-3 h-3 bg-emerald-500 rounded"></span>
        <span>Satisfied (&lt;0.3)</span>
    </div>
    <div className="flex items-center space-x-1.5">
        <span className="w-3 h-3 bg-yellow-400 rounded"></span>
        <span>Low Gap (0.4 - 0.8)</span>
    </div>
    <div className="flex items-center space-x-1.5">
        <span className="w-3 h-3 bg-orange-500 rounded"></span>
        <span>Medium Gap (0.9 - 1.5)</span>
    </div>
    <div className="flex items-center space-x-1.5">
        <span className="w-3 h-3 bg-red-600 rounded"></span>
        <span>Critical Gap (&gt;1.5)</span>
    </div>
</div>
</div>
);
}

```

Slidebar.tsx.txt

"use client";

import React from 'react';

import {

LayoutDashboard, Award, Compass, Sparkles, FileCheck, BookOpen, User, BarChart3, Settings, ShieldAlert, Layers

} from 'lucide-react';

interface SidebarProps {

activeTab: string;

setActiveTab: (tab: string) => void;

roleMode: 'learner' | 'admin';

}

interface NavItem {

id: string;

label: string;

icon: any;

badge?: string;

}

export default function Sidebar({ activeTab, setActiveTab, roleMode }: SidebarProps) {

const learnerNav: NavItem[] = [

{ id: 'dashboard', label: 'Dashboard', icon: LayoutDashboard },

{ id: 'competencies', label: 'My Competencies', icon: Award },

{ id: 'recommendations', label: 'Learning Recommendations', icon: BookOpen },

{ id: 'navigator', label: 'Career Navigator', icon: Compass, badge: 'Unique' },

{ id: 'assistant', label: 'AI Assistant', icon: Sparkles, badge: 'RAG' },

{ id: 'assessments', label: 'Assessments', icon: FileCheck },

{ id: 'my-learning', label: 'My Learning', icon: Layers }

];

const adminNav: NavItem[] = [

{ id: 'admin-overview', label: 'Overview', icon: BarChart3 },

{ id: 'admin-heatmap', label: 'Workforce Competencies', icon: Award },

{ id: 'admin-gaps', label: 'Skill Gaps', icon: ShieldAlert },

{ id: 'admin-assessments', label: 'Assessment Generator', icon: FileCheck },

```

    { id: 'admin-rules', label: 'Rules & Framework', icon: Settings }
  ];

  const navItems = roleMode === 'admin' ? adminNav : learnerNav;

  return (
    <aside className="w-64 bg-slate-900 border-r border-slate-800 text-slate-300 min-h-[
    calc(100vh-57px)] p-4 flex flex-col justify-between">
      <div className="space-y-6">
        <div className="px-3 py-2 text-[11px] font-bold uppercase tracking-wider text-slate-
        500">
          {roleMode === 'admin' ? 'Government Admin Panel' : 'Officer Portal'}
        </div>

        <nav className="space-y-1">
          {navItems.map((item) => {
            const Icon = item.icon;
            const isActive = activeTab === item.id;
            return (
              <button
                key={item.id}
                onClick={() => setActiveTab(item.id)}
                className={`w-full flex items-center justify-between px-3.5 py-2.5 rounded-xl tex
t-xs font-semibold transition ${
                  isActive
                    ? 'bg-blue-600 text-white shadow-md shadow-blue-900/50'
                    : 'text-slate-400 hover:text-white hover:bg-slate-800/70'
                }`}
              >
                <div className="flex items-center space-x-3">
                  <Icon className={`w-4 h-4 ${isActive ? 'text-white' : 'text-slate-400'}`} />
                  <span>{item.label}</span>
                </div>
                  {item.badge && (
                    <span className={`text-[9px] font-bold uppercase px-1.5 py-0.5 rounded-md $
                    {
                      isActive ? 'bg-white text-blue-700' : 'bg-blue-500/20 text-blue-400 border border
                    -blue-400/30'

```

```

    }}>
    {item.badge}
  </span>
)}
</button>
);
}}
</nav>
</div>

```

```

<div className="pt-4 border-t border-slate-800 text-center">
  <div className="p-3 bg-slate-800/60 rounded-xl border border-slate-700/50">
    <span className="text-[10px] text-slate-400 block font-medium">MoSPI Competen
cy Model</span>
    <span className="text-xs font-bold text-white block mt-0.5">NSSTA Certified v1.2<
/span>
  </div>
</div>
</aside>
);
}

```

ExplainabilityModal,tsx.txt

"use client";

```

import React from 'react';
import { X, CheckCircle2, AlertTriangle, ArrowRight, ShieldCheck, Info } from 'lucide-react';

```

```

interface ExplainabilityModalProps {
  isOpen: boolean;
  onClose: () => void;
  data: any;
}

```

```

export default function ExplainabilityModal({ isOpen, onClose, data }: ExplainabilityModa
lProps) {
  if (!isOpen || !data) return null;

  const {

```

```

    learner_name,
    current_role,
    target_role,
    competency_name,
    domain,
    assessed_level,
    required_level,
    gap,
    severity,
    course_title,
    course_provider,
    course_duration,
    coverage_level,
    recommendation_reason,
    score,
    score_breakdown
  } = data;

  return (
    <div className="fixed inset-0 z-50 flex items-center justify-center bg-slate-900/60 backdrop-blur-sm p-4 animate-fade-in">
      <div className="bg-white rounded-2xl shadow-2xl border border-slate-200 w-full max-w-3xl overflow-hidden max-h-[90vh] flex flex-col">

        { /* Header */ }
        <div className="bg-slate-900 text-white px-6 py-4 flex items-center justify-between">
          >
          <div className="flex items-center space-x-3">
            <div className="p-2 bg-blue-600 rounded-lg">
              <ShieldCheck className="w-5 h-5 text-white" />
            </div>
            <div>
              <h2 className="text-lg font-bold">Competency Explainability Trail</h2>
              <p className="text-xs text-slate-300">Auditable, evidence-based recommendation reasoning</p>
            </div>
          </div>
          <button

```

```

onClick={onClose}
className="p-1.5 text-slate-400 hover:text-white hover:bg-slate-800 rounded-lg transition"
>
  <X className="w-5 h-5" />
</button>
</div>

{/* Content */}
<div className="p-6 overflow-y-auto space-y-6">

  {/* Course Badge */}
  <div className="bg-blue-50 border border-blue-200 rounded-xl p-4 flex items-center justify-between">
    <div>
      <span className="text-xs font-semibold text-blue-700 uppercase tracking-wider">Recommended Course</span>
      <h3 className="text-xl font-bold text-slate-900">{course_title}</h3>
      <p className="text-sm text-slate-600">Provider: {course_provider} | Duration: {course_duration}</p>
    </div>
    <div className="text-right">
      <span className="text-2xl font-black text-blue-700">{score}/100</span>
      <p className="text-xs text-slate-500 font-medium">Relevance Score</p>
    </div>
  </div>

  {/* Visual Trail Flow (Image 3 layout) */}
  <div className="border border-slate-200 rounded-xl p-5 bg-slate-50">
    <h4 className="text-xs font-bold text-slate-500 uppercase tracking-wider mb-4 flex items-center gap-1.5">
      <Info className="w-4 h-4 text-blue-600" /> Audit Trail (Why this course?)
    </h4>

    <div className="grid grid-cols-5 gap-2 text-center text-xs">
      <div className="p-3 bg-white rounded-lg border border-slate-200 shadow-sm">
        <span className="text-slate-400 font-medium block mb-1">Profile</span>
        <span className="font-bold text-slate-800 block text-ellipsis overflow-hidden">{l

```

```
earner_name}</span>
    <span className="text-[10px] text-slate-500">{current_role}</span>
</div>

<div className="p-3 bg-white rounded-lg border border-slate-200 shadow-sm">
    <span className="text-slate-400 font-medium block mb-1">Assessed Level</span>
    <span className="font-bold text-slate-800 block text-base">{assessed_level}</span>
    <span className="text-[10px] text-slate-500">Current Capability</span>
</div>

<div className="p-3 bg-white rounded-lg border border-slate-200 shadow-sm">
    <span className="text-slate-400 font-medium block mb-1">Required Level</span>
    <span className="font-bold text-slate-800 block text-base">{required_level}</span>
    <span className="text-[10px] text-slate-500">Role Benchmark</span>
</div>

<div className="p-3 bg-red-50 rounded-lg border border-red-200 shadow-sm">
    <span className="text-red-600 font-medium block mb-1">Verified Gap</span>
    <span className="font-bold text-red-700 block text-base">{gap} Level{gap > 1 ?
's' : ''}</span>
    <span className="text-[10px] text-red-600 font-bold uppercase">{severity}</span>
</div>

<div className="p-3 bg-emerald-50 rounded-lg border border-emerald-200 shadow-sm">
    <span className="text-emerald-600 font-medium block mb-1">Course Coverage</span>
    <span className="font-bold text-emerald-800 block text-base">Level {coverage_level}</span>
    <span className="text-[10px] text-emerald-600 font-semibold">Verified Match</span>
</div>
</div>
```

</div>

{/\* Reasoning Text \*/}

<div className="bg-slate-900 text-white rounded-xl p-4 space-y-2">

<h4 className="text-xs font-bold text-blue-400 uppercase tracking-wider">Deterministic Evidence Reasoning</h4>

<p className="text-sm leading-relaxed text-slate-200">

"{recommendation\_reason}"

</p>

</div>

{/\* Score Breakdown Bars \*/}

<div className="space-y-3">

<h4 className="text-xs font-bold text-slate-500 uppercase tracking-wider">Score Breakdown Metrics</h4>

<div className="grid grid-cols-5 gap-3">

<div className="bg-slate-50 border border-slate-200 p-3 rounded-lg text-center">

<span className="text-xs text-slate-500 block">Gap Match</span>

<span className="text-lg font-bold text-blue-700">{score\_breakdown.gap\_match}/40</span>

</div>

<div className="bg-slate-50 border border-slate-200 p-3 rounded-lg text-center">

<span className="text-xs text-slate-500 block">Role Relevance</span>

<span className="text-lg font-bold text-blue-700">{score\_breakdown.target\_role}/25</span>

</div>

<div className="bg-slate-50 border border-slate-200 p-3 rounded-lg text-center">

<span className="text-xs text-slate-500 block">Coverage</span>

<span className="text-lg font-bold text-blue-700">{score\_breakdown.competency\_coverage}/15</span>

</div>

<div className="bg-slate-50 border border-slate-200 p-3 rounded-lg text-center">

<span className="text-xs text-slate-500 block">Preference</span>

<span className="text-lg font-bold text-blue-700">{score\_breakdown.preference}</span>

```
e}/10</span>
    </div>

    <div className="bg-slate-50 border border-slate-200 p-3 rounded-lg text-center">
      <span className="text-xs text-slate-500 block">Quality</span>
      <span className="text-lg font-bold text-blue-700">{score_breakdown.quality}/1
0</span>
    </div>
  </div>
</div>

{ /* Footer */ }
<div className="bg-slate-50 border-t border-slate-200 px-6 py-3 flex items-center ju
stify-between text-xs text-slate-500">
  <span>Rules Version: 1.2.0 | MoSPI Framework Standard</span>
  <button
    onClick={onClose}
    className="px-4 py-2 bg-slate-900 text-white rounded-lg font-medium hover:bg-sl
ate-800 transition"
  >
    Close Trail
  </button>
</div>

</div>
</div>
);
}
```

Scr/lib/api

```
const API_BASE = process.env.NEXT_PUBLIC_API_URL || "http://localhost:8000/api";

export async function fetchApi<T>(endpoint: string, options?: RequestInit): Promise<T> {
  const res = await fetch(`${API_BASE}${endpoint}`, {
    headers: {
      "Content-Type": "application/json",
```

```

    ...(options?.headers || {})
  },
  ...options
});

if (!res.ok) {
  throw new Error(`API Error ${res.status}: ${res.statusText}`);
}

return res.json();
}

export const api = {
  getDemoUser: () => fetchApi<any>("/auth/demo-user"),
  getLearnerGaps: (learnerId: number, targetRoleId?: number) =>
    fetchApi<any[]>(`/learners/${learnerId}/gaps${targetRoleId ? `?target_role_id=${targetRoleId}` : ""}`),
  getRecommendations: (learnerId: number) => fetchApi<any[]>(`/recommendations/${learnerId}`),
  getCareerNavigator: (learnerId: number, targetRoleId?: number) =>
    fetchApi<any>(`/career-navigator/${learnerId}${targetRoleId ? `?target_role_id=${targetRoleId}` : ""}`),
  getRoles: () => fetchApi<any[]>("/roles"),
  getCourses: () => fetchApi<any[]>("/courses"),
  chatAssistant: (learnerId: number, message: string) =>
    fetchApi<any>("/assistant/chat", {
      method: "POST",
      body: JSON.stringify({ learner_id: learnerId, message })
    }),
  generateQuiz: (docId: number) =>
    fetchApi<any>(`/assessments/${docId}/generate`, { method: "POST" }),
  getQuizQuestions: (docId: number) => fetchApi<any[]>(`/assessments/${docId}/questions`),
  publishQuiz: (docId: number) => fetchApi<any>(`/assessments/${docId}/publish`, { method: "POST" }),
  getAdminAnalytics: () => fetchApi<any>("/admin/analytics");
};

```

page.tsx.txt

"use client";

```
import React, { useState, useEffect } from 'react';
import Navbar from '@components/Navbar';
import Sidebar from '@components/Sidebar';
import CompetencyRadarChart from '@components/CompetencyRadarChart';
import ExplainabilityModal from '@components/ExplainabilityModal';
import AdminHeatmap from '@components/AdminHeatmap';
import { api } from '@lib/api';
import {
  ShieldAlert, BookOpen, Compass, Sparkles, Award, ArrowRight, CheckCircle2,
  AlertCircle, Upload, FileText, Send, Check, Play, FileCheck, Layers, RefreshCw, BarChart2
} from 'lucide-react';
import { BarChart, Bar, XAxis, YAxis, Tooltip, ResponsiveContainer, PieChart, Pie, Cell } from 'recharts';
```

```
export default function Home() {
  const [roleMode, setRoleMode] = useState<'learner' | 'admin'>('learner');
  const [activeTab, setActiveTab] = useState<string>('dashboard');

  // Data states
  const [user, setUser] = useState<any>(null);
  const [gaps, setGaps] = useState<any[]>([]);
  const [recommendations, setRecommendations] = useState<any[]>([]);
  const [careerNav, setCareerNav] = useState<any>(null);
  const [selectedTargetRole, setSelectedTargetRole] = useState<number>(2);
  const [roles, setRoles] = useState<any[]>([]);
  const [analytics, setAnalytics] = useState<any>(null);

  // Modal state
  const [selectedExplainability, setSelectedExplainability] = useState<any>(null);
  const [isExplainModalOpen, setIsExplainModalOpen] = useState<boolean>(false);

  // Chat state
  const [chatMessages, setChatMessages] = useState<any[]>([
    {
```

```

    sender: 'assistant',
    text: 'Namaste Ananya Sharma! I am your MoSPI AI Learning Advisor. How can I assist you with your competency gaps or target role requirements today?',
    citations: []
  }
]);
const [chatInput, setChatInput] = useState<string>("");
const [isChatLoading, setIsChatLoading] = useState<boolean>(false);

// Quiz state
const [quizDoc, setQuizDoc] = useState<any>(null);
const [quizQuestions, setQuizQuestions] = useState<any[]>([]);
const [isGeneratingQuiz, setIsGeneratingQuiz] = useState<boolean>(false);

useEffect(() => {
  loadInitialData();
}, []);

const loadInitialData = async () => {
  try {
    const demoUser = await api.getDemoUser();
    setUser(demoUser);
    setSelectedTargetRole(demoUser.target_role_id || 2);

    const [gapRes, recRes, navRes, roleRes, adminRes] = await Promise.all([
      api.getLearnerGaps(demoUser.id),
      api.getRecommendations(demoUser.id),
      api.getCareerNavigator(demoUser.id, demoUser.target_role_id || 2),
      api.getRoles(),
      api.getAdminAnalytics()
    ]);

    setGaps(gapRes);
    setRecommendations(recRes);
    setCareerNav(navRes);
    setRoles(roleRes);
    setAnalytics(adminRes);
  } catch (err) {

```

```
    console.error("Failed loading data", err);
  }
};
```

```
const handleRoleChange = async (targetId: number) => {
  setSelectedTargetRole(targetId);
  if (user) {
    const [gapRes, navRes] = await Promise.all([
      api.getLearnerGaps(user.id, targetId),
      api.getCareerNavigator(user.id, targetId)
    ]);
    setGaps(gapRes);
    setCareerNav(navRes);
  }
};
```

```
const handleOpenExplainability = (rec: any) => {
  setSelectedExplainability(rec.explainability);
  setIsExplainModalOpen(true);
};
```

```
const handleSendChatMessage = async (msgText?: string) => {
  const textToSend = msgText || chatInput;
  if (!textToSend.trim() || !user) return;
```

```
  const userMsg = { sender: 'user', text: textToSend };
  setChatMessages(prev => [...prev, userMsg]);
  if (!msgText) setChatInput("");
  setIsChatLoading(true);
```

```
  try {
    const resp = await api.chatAssistant(user.id, textToSend);
    setChatMessages(prev => [...prev, {
      sender: 'assistant',
      text: resp.answer,
      citations: resp.citations,
      mode: resp.mode
    }]);
```

```

} catch (err) {
  setChatMessages(prev => [...prev, {
    sender: 'assistant',
    text: "I couldn't find enough verified information to answer that question.",
    citations: []
  }]);
} finally {
  setIsChatLoading(false);
}
};

```

```

const handleGenerateQuizSample = async () => {
  setIsGeneratingQuiz(true);
  try {
    const res = await api.generateQuiz(1);
    setQuizQuestions(res.questions);
  } catch (err) {
    console.error(err);
  } finally {
    setIsGeneratingQuiz(false);
  }
};

```

```

const COLORS = ['#2563eb', '#3b82f6', '#60a5fa', '#93c5fd', '#cbd5e1'];

```

```

return (
  <div className="min-h-screen bg-slate-100 flex flex-col font-sans">
    {/* Top Navbar */}
    <Navbar
      user={user}
      currentRoleMode={roleMode}
      onToggleRoleMode={() => setRoleMode(roleMode === 'learner' ? 'admin' : 'learner')}
    />

    <div className="flex flex-1">
      {/* Navigation Sidebar */}
      <Sidebar
        activeTab={activeTab}

```

```

    setActiveTab={setActiveTab}
    roleMode={roleMode}
  />

  { /* Main Content Area */ }
  <main className="flex-1 p-6 overflow-y-auto max-w-7xl">

    { /* =====
===== */ }
    { /* 1. LEARNER DASHBOARD (Image Panel 1 Layout) */ }
    { /* =====
===== */ }
    {roleMode === 'learner' && activeTab === 'dashboard' && (
      <div className="space-y-6">

        { /* Header Welcome Bar */ }
        <div className="bg-white rounded-2xl p-6 border border-slate-200 shadow-sm fl
ex items-center justify-between">
          <div>
            <h1 className="text-2xl font-black text-slate-900 flex items-center gap-2">
              Welcome, {user?.name || 'Ananya Sharma'}! <span className="text-xl">👋</sp
an>
            </h1>
            <p className="text-sm text-slate-500 font-medium mt-1">
              {user?.designation || 'Statistical Officer'} | {user?.department || 'Demo Statistics
Department'}
            </p>
          </div>

          <div className="flex items-center space-x-4 bg-slate-50 border border-slate-200
px-4 py-2.5 rounded-xl">
            <div>
              <span className="text-[11px] font-bold text-slate-400 uppercase tracking-wid
er block">Target Role</span>
              <span className="text-sm font-bold text-slate-800">
                {roles.find(r => r.id === selectedTargetRole)?.name || 'Senior Statistical Officer'
              }
            </span>

```

```
    </div>
    <button
      onClick={() => setActiveTab('navigator')}
      className="px-3 py-1.5 bg-blue-600 hover:bg-blue-700 text-white rounded-lg text-xs font-semibold transition"
    >
      Change Target Role
    </button>
  </div>
</div>
```

```
{/* Grid Layout: Competency Radar vs Top Skill Gaps */}
<div className="grid grid-cols-1 md:grid-cols-2 gap-6">

  {/* Competency Overview Radar Chart */}
  <div className="bg-white rounded-2xl p-6 border border-slate-200 shadow-sm flex flex-col justify-between">
    <div className="flex items-center justify-between mb-2">
      <h2 className="text-base font-bold text-slate-900">Competency Overview</h2>
      <span className="text-xs text-slate-400 font-medium">6 Framework Domains</span>
    </div>

    <CompetencyRadarChart />
  </div>

  </div>
```

```
{/* Top Skill Gaps List */}
<div className="bg-white rounded-2xl p-6 border border-slate-200 shadow-sm flex flex-col justify-between">
  <div className="flex items-center justify-between mb-4">
    <h2 className="text-base font-bold text-slate-900">Top Skill Gaps</h2>
    <button onClick={() => setActiveTab('competencies')} className="text-xs font-bold text-blue-600 hover:underline">
      View all competencies →
    </button>
  </div>
```

```

<div className="space-y-3.5 flex-1">
  {gaps.filter(g => g.gap > 0).slice(0, 3).map((gapItem, idx) => (
    <div key={idx} className="flex items-center justify-between p-3.5 bg-slate-50
rounded-xl border border-slate-200/80">
      <div className="flex items-center space-x-3">
        <span className="font-extrabold text-slate-400 text-sm">{idx + 1}.</span>
        <div>
          <h3 className="text-sm font-bold text-slate-900">{gapItem.competency_
name}</h3>
          <p className="text-xs text-slate-500">Gap: {gapItem.gap} Level{gapItem.
gap > 1 ? 's': ''} (Assessed: {gapItem.assessed_level} vs Req: {gapItem.required_level})</p>
        </div>
      </div>
    </div>

    <span className={`px-3 py-1 rounded-full text-xs font-bold ${
      gapItem.severity === 'Critical' ? 'bg-red-100 text-red-700 border border-red-2
00':
      gapItem.severity === 'Medium' ? 'bg-orange-100 text-orange-700 border bor
der-orange-200':
      'bg-emerald-100 text-emerald-700 border border-emerald-200'
    }`}>
      {gapItem.severity}
    </span>
  </div>
  </div>
  </div>
  </div>

  {/* Recommended for You Grid */}
  <div className="bg-white rounded-2xl p-6 border border-slate-200 shadow-sm s
pace-y-4">
    <div className="flex items-center justify-between">
      <div>
        <h2 className="text-lg font-bold text-slate-900">Recommended for You</h2>
        <p className="text-xs text-slate-500">Grounded in verified skill gaps and curr

```

```
ent catalogue</p>
  </div>
  <button onClick={() => setActiveTab('recommendations')} className="text-xs font-bold text-blue-600 hover:underline">
    Browse full catalogue →
  </button>
</div>

<div className="grid grid-cols-1 md:grid-cols-2 gap-4">
  {recommendations.slice(0, 2).map((rec, idx) => (
    <div key={idx} className="border border-slate-200 rounded-xl p-5 hover:shadow-md transition bg-slate-50/50 flex flex-col justify-between">
      <div>
        <div className="flex items-start justify-between">
          <div>
            <span className="text-[11px] font-bold text-blue-700 uppercase bg-blue-100 px-2 py-0.5 rounded">
              Score: {rec.score}/100
            </span>
            <h3 className="text-base font-bold text-slate-900 mt-2">{rec.course_title}
          </h3>
            <p className="text-xs text-slate-500 mt-1">Provider: {rec.provider} | Duration: {rec.duration}</p>
          </div>
          </div>
        </div>
        <p className="text-xs text-slate-600 mt-3 line-clamp-2 bg-white p-2.5 rounded-lg border border-slate-200">
          "{rec.explainability.recommendation_reason}"
        </p>
      </div>

      <div className="mt-4 pt-3 border-t border-slate-200 flex items-center justify-between">
        <button
          onClick={() => handleOpenExplainability(rec)}
          className="text-xs font-bold text-blue-600 hover:text-blue-800 flex items-center space-x-1">
          >
```

```

        <span>Why this course?</span>
        <ArrowRight className="w-3.5 h-3.5" />
    </button>
    <button className="px-3 py-1.5 bg-slate-900 text-white rounded-lg text-xs font-semibold hover:bg-slate-800 transition">
        Enroll now
    </button>
</div>
</div>
)}}
</div>
</div>

</div>
)}

{ /* =====
===== */}

{ /* 2. CAREER NAVIGATOR (Unique Feature 2 - Image Panel 4 Layout) */}
{ /* =====
===== */}

{ (roleMode === 'learner' && activeTab === 'navigator') && (
    <div className="space-y-6">

        { /* Top Banner */}
        <div className="bg-slate-900 text-white rounded-2xl p-6 border border-slate-800 shadow-md">
            <div className="flex items-center justify-between">
                <div>
                    <span className="text-xs font-bold text-blue-400 uppercase tracking-wider block">Career & Role Navigator</span>
                    <h1 className="text-2xl font-extrabold mt-1">Plan your learning for your target role</h1>
                    <p className="text-xs text-slate-300 mt-1">Objective competency readiness matrix for government career progression</p>
                </div>
            </div>
        </div>
    </div>

```

```

        <div className="bg-slate-800 p-3 rounded-xl border border-slate-700 flex items
-center space-x-4">
        <div className="text-right">
        <span className="text-xs text-slate-400 block font-medium">Select Target R
ole</span>
        <select
        value={selectedTargetRole}
        onChange={(e) => handleRoleChange(Number(e.target.value))}
        className="bg-slate-900 text-white border border-slate-700 text-sm font-bo
ld rounded-lg px-3 py-1.5 focus:outline-none focus:border-blue-500"
        >
        {roles.map(r => (
        <option key={r.id} value={r.id}>{r.name}</option>
        )))}
        </select>
        </div>
        </div>
        </div>
        </div>
    </div>

```

```

    {/* Competency Readiness Gauge & Role Requirements Table */}
    <div className="bg-white rounded-2xl p-6 border border-slate-200 shadow-sm s
pace-y-4">
        <div className="flex items-center justify-between border-b border-slate-200 pb-
4">
            <div>
                <h2 className="text-lg font-bold text-slate-900">Competency Gap for Target
Role</h2>
                <p className="text-xs text-slate-500">Benchmark requirement comparison vs
assessed capability</p>
            </div>
            <div className="text-right">
                <span className="text-2xl font-black text-blue-700">{careerNav?.competency
_readiness_pct || 78}%</span>
                <span className="text-xs text-slate-500 block font-medium">Competency Re
adiness</span>
            </div>
        </div>
    </div>

```

```
<div className="overflow-x-auto">
  <table className="w-full text-xs text-left border-collapse">
    <thead>
      <tr className="bg-slate-50 text-slate-700 uppercase font-bold border-b border-slate-200">
        <th className="p-3">Competency</th>
        <th className="p-3">Domain</th>
        <th className="p-3 text-center">Required Level</th>
        <th className="p-3 text-center">Current Level</th>
        <th className="p-3 text-center">Gap</th>
        <th className="p-3 text-center">Priority</th>
      </tr>
    </thead>
    <tbody className="divide-y divide-slate-200">
      {careerNav?.priority_gaps?.map((gapItem: any, idx: number) => (
        <tr key={idx} className="hover:bg-slate-50">
          <td className="p-3 font-bold text-slate-900">{gapItem.competency_name}</td>

          <td className="p-3 text-slate-500">{gapItem.domain}</td>
          <td className="p-3 text-center font-bold text-slate-800">{gapItem.required_level}</td>
          <td className="p-3 text-center text-slate-600">{gapItem.current_level}</td>
          >
            <td className="p-3 text-center">
              <span className={`px-2 py-0.5 rounded text-xs font-bold ${
                gapItem.gap > 0 ? 'bg-red-100 text-red-700' : 'bg-emerald-100 text-emerald-700'
              }`}>
                {gapItem.gap}
              </span>
            </td>
            <td className="p-3 text-center">
              <span className="uppercase text-[10px] font-bold px-2 py-0.5 rounded bg-slate-100 text-slate-700">
                {gapItem.priority}
              </span>
            </td>
          </td>
        </tr>
      )
    )
  }
</tbody>
</table>
</div>
```

```
        </tr>
      )}}
    </tbody>
  </table>
</div>
</div>
```

```
{/* Sequenced Learning Path (Step 1..N) */}
<div className="bg-white rounded-2xl p-6 border border-slate-200 shadow-sm space-y-4">
  <h2 className="text-lg font-bold text-slate-900">Recommended Learning Path Sequence</h2>
  <p className="text-xs text-slate-500">Ordered sequence to address target role competency requirements</p>

  <div className="space-y-4 pt-2">
    {careerNav?.recommended_path?.map((step: any, idx: number) => (
      <div key={idx} className="flex items-start space-x-4 p-4 rounded-xl border border-slate-200 bg-slate-50/70 hover:border-blue-300 transition">
        <div className="w-8 h-8 rounded-full bg-blue-600 text-white flex items-center justify-center font-bold text-sm shrink-0 mt-0.5">
          {step.step_number}
        </div>

        <div className="flex-1">
          <div className="flex items-center justify-between">
            <h3 className="text-base font-bold text-slate-900">{step.course_title}</h3>
            <span className="text-xs font-semibold text-slate-500">{step.duration}</span>
          </div>

          <p className="text-xs text-slate-600 mt-1 font-medium">{step.reason}</p>
          {step.prerequisites && (
            <span className="text-[11px] text-blue-600 font-semibold block mt-2">
              Prerequisite: {step.prerequisites}
            </span>
          )}
        </div>
      </div>
    )}
  </div>
</div>
```

```
    </div>
  )))
</div>
</div>
```

```
</div>
})
```

```
    { /* =====
===== */ }
```

```
    { /* 3. AI ASSISTANT (RAG Powered - Image Panel 5 Layout) */ }
```

```
    { /* =====
===== */ }
```

```
    {(roleMode === 'learner' && activeTab === 'assistant') && (
      <div className="h-[calc(100vh-100px)] flex flex-col bg-white rounded-2xl border border-slate-200 shadow-sm overflow-hidden">
```

```
      { /* Chat Header */ }
      <div className="bg-slate-900 text-white p-4 flex items-center justify-between border-b border-slate-800">
```

```
        <div className="flex items-center space-x-3">
          <div className="p-2 bg-blue-600 rounded-lg">
            <Sparkles className="w-5 h-5 text-white" />
          </div>
```

```
        <div>
          <h2 className="text-base font-bold">MoSPI AI Learning Advisor</h2>
          <p className="text-xs text-slate-400">Context-Aware & Grounded in Official Frameworks</p>
```

```
        </div>
      </div>
      <span className="text-[11px] bg-emerald-500/20 text-emerald-300 border border-emerald-500/30 px-2.5 py-1 rounded-full font-semibold">
```

```
        RAG Active
```

```
      </span>
    </div>
```

```
    { /* Suggested Prompt Pills */ }
```

```
    <div className="bg-slate-50 p-3 border-b border-slate-200 flex flex-wrap gap-2">
```

```
{[
  "Why was Advanced Data Analysis recommended to me?",
  "What competencies do I need for Senior Statistical Officer?",
  "What should I learn next?",
  "Explain my Data Analysis gap."
].map((prompt, idx) => (
  <button
    key={idx}
    onClick={() => handleSendChatMessage(prompt)}
    className="text-xs bg-white text-slate-700 border border-slate-300 hover:bor
der-blue-500 hover:text-blue-700 px-3 py-1.5 rounded-full transition shadow-2xs font-me
dium"
  >
    "{prompt}"
  </button>
  )))
</div>
```

```
{/* Chat History */}
<div className="flex-1 p-6 overflow-y-auto space-y-4">
  {chatMessages.map((msg, idx) => (
    <div key={idx} className={`flex flex-col ${msg.sender === 'user' ? 'items-end' : 'it
ems-start'}`}>
      <div className={`max-w-2xl rounded-2xl p-4 text-sm ${
        msg.sender === 'user'
          ? 'bg-blue-600 text-white rounded-br-none'
          : 'bg-slate-100 text-slate-800 border border-slate-200 rounded-bl-none'
        }`}>
        <p className="whitespace-pre-wrap leading-relaxed">{msg.text}</p>

        {/* Source Citations */}
        {msg.citations && msg.citations.length > 0 && (
          <div className="mt-3 pt-2 border-t border-slate-300/50 space-y-1">
            <span className="text-[10px] font-bold text-slate-500 uppercase tracking-
wider block">Grounded Sources:</span>
            {msg.citations.map((cite: any, cldx: number) => (
              <div key={cldx} className="bg-white/80 p-2 rounded text-[11px] text-slate
-700 border border-slate-200">
```

```

        <span className="font-bold text-blue-700">{cite.title}</span> - {cite.clau
se_or_source}
    </div>
    )))
</div>
})
</div>
</div>
)))

{isChatLoading && (
  <div className="flex items-center space-x-2 text-slate-400 text-xs italic">
    <RefreshCw className="w-4 h-4 animate-spin" />
    <span>Retrieving context & synthesizing grounded answer...</span>
  </div>
)}
</div>

{/* Chat Input Bar */}
<div className="p-4 border-t border-slate-200 bg-slate-50 flex items-center spac
e-x-2">
  <input
    type="text"
    value={chatInput}
    onChange={(e) => setChatInput(e.target.value)}
    onKeyDown={(e) => e.key === 'Enter' && handleSendChatMessage()}
    placeholder="Ask a question about your competency gaps, target role, or recom
mendations..."
    className="flex-1 bg-white border border-slate-300 rounded-xl px-4 py-2.5 text
-sm focus:outline-none focus:border-blue-500 text-slate-800"
  />
  <button
    onClick={() => handleSendChatMessage()}
    disabled={isChatLoading}
    className="p-2.5 bg-blue-600 hover:bg-blue-700 text-white rounded-xl transiti
on disabled:opacity-50"
  >
    <Send className="w-5 h-5" />

```

```
        </button>
    </div>

</div>
)}

{ /* =====
===== */}
{ /* 4. ADMIN DASHBOARD (Image Panel 7 Layout) */}
{ /* =====
===== */}
{roleMode === 'admin' && (
  <div className="space-y-6">

    { /* Executive KPI Stats Cards */}
    <div className="grid grid-cols-4 gap-4">
      <div className="bg-white p-5 rounded-2xl border border-slate-200 shadow-sm flex items-center space-x-4">
        <div className="p-3 bg-blue-100 text-blue-700 rounded-xl">
          <BarChart2 className="w-6 h-6" />
        </div>
        <div>
          <span className="text-2xl font-black text-slate-900">{analytics?.total_officials || 248}</span>
          <span className="text-xs text-slate-500 block font-medium">Total Officials</span>
        </div>
      </div>

      <div className="bg-white p-5 rounded-2xl border border-slate-200 shadow-sm flex items-center space-x-4">
        <div className="p-3 bg-red-100 text-red-700 rounded-xl">
          <ShieldAlert className="w-6 h-6" />
        </div>
        <div>
          <span className="text-2xl font-black text-slate-900">{analytics?.critical_skill_gaps || 36}</span>
          <span className="text-xs text-slate-500 block font-medium">Critical Skill Gap
```

```
s</span>
    </div>
</div>

    <div className="bg-white p-5 rounded-2xl border border-slate-200 shadow-sm
flex items-center space-x-4">
    <div className="p-3 bg-indigo-100 text-indigo-700 rounded-xl">
        <BookOpen className="w-6 h-6" />
    </div>
    <div>
        <span className="text-2xl font-black text-slate-900">{analytics?.courses_enr
olled || 412}</span>
        <span className="text-xs text-slate-500 block font-medium">Courses Enrolle
d</span>
    </div>
</div>

    <div className="bg-white p-5 rounded-2xl border border-slate-200 shadow-sm
flex items-center space-x-4">
    <div className="p-3 bg-emerald-100 text-emerald-700 rounded-xl">
        <CheckCircle2 className="w-6 h-6" />
    </div>
    <div>
        <span className="text-2xl font-black text-slate-900">{analytics?.completion_r
ate_pct || 68}%</span>
        <span className="text-xs text-slate-500 block font-medium">Completion Rate
</span>
    </div>
</div>

    </div>
</div>

    {/* Organization Competency Heatmap */}
    <div className="bg-white rounded-2xl p-6 border border-slate-200 shadow-sm s
pace-y-4">
    <div className="flex items-center justify-between">
    <div>
        <h2 className="text-lg font-bold text-slate-900">Organization Competency H
eatmap</h2>
```

```

    <p className="text-xs text-slate-500">Average competency gap severity by rol
e and framework domain</p>
  </div>
</div>

  <AdminHeatmap data={analytics?.heatmap || []} />
</div>

  {/* Charts Grid: Skill Gaps Bar Chart vs Course Demand Pie Chart */}
  <div className="grid grid-cols-2 gap-6">

    {/* Top Skill Gaps Across Organization */}
    <div className="bg-white rounded-2xl p-6 border border-slate-200 shadow-sm"
>
      <h2 className="text-base font-bold text-slate-900 mb-4">Top 5 Skill Gaps Acro
ss Organization</h2>
      <div className="h-64">
        <ResponsiveContainer width="100%" height="100%">
          <BarChart data={analytics?.top_skill_gaps || []} layout="vertical">
            <XAxis type="number" />
            <YAxis dataKey="name" type="category" width={140} tick={{ fontSize: 11 }} /
>
              <Tooltip />
              <Bar dataKey="count" fill="#2563eb" radius={[0, 6, 6, 0]} />
            </BarChart>
          </ResponsiveContainer>
        </div>
      </div>

      {/* Course Demand */}
      <div className="bg-white rounded-2xl p-6 border border-slate-200 shadow-sm"
>
        <h2 className="text-base font-bold text-slate-900 mb-4">Course Demand Shar
e</h2>
        <div className="h-64 flex items-center justify-center">
          <ResponsiveContainer width="100%" height="100%">
            <PieChart>
              <Pie
```

```

      data={analytics?.course_demand || []}
      cx="50%"
      cy="50%"
      innerRadius={60}
      outerRadius={90}
      paddingAngle={3}
      dataKey="value"
    >
      {analytics?.course_demand?.map((entry: any, index: number) => (
        <Cell key={`cell-${index}`} fill={COLORS[index % COLORS.length]} />
      ))}
    </Pie>
    <Tooltip />
  </PieChart>
</ResponsiveContainer>
</div>
</div>

```

```

</div>

```

```

</div>

```

```

  )}

```

```

    {/* =====
===== */}

```

```

    {/* 5. OTHER TABS (My Competencies, Learning Recommendations, Assessments)
*/}

```

```

    {/* =====
===== */}

```

```

    {roleMode === 'learner' && activeTab === 'competencies' && (
      <div className="bg-white rounded-2xl p-6 border border-slate-200 shadow-sm space-y-4">
        <h1 className="text-xl font-bold text-slate-900">My Competency Profile</h1>
        <p className="text-xs text-slate-500">Official capability assessment levels across 6 framework domains</p>

```

```

      <div className="grid grid-cols-2 gap-4 pt-4">
        {gaps.map((g, idx) => (

```

```

    <div key={idx} className="p-4 border border-slate-200 rounded-xl bg-slate-50 flex items-center justify-between">
      <div>
        <span className="text-[10px] uppercase font-bold text-slate-400">{g.domain}</span>
        <h3 className="text-base font-bold text-slate-900">{g.competency_name}</h3>
        <p className="text-xs text-slate-500 mt-1">Assessed Level: {g.assessed_level} / Required: {g.required_level}</p>
      </div>

      <span className={`px-3 py-1 rounded-full text-xs font-bold ${g.gap === 0 ? 'bg-emerald-100 text-emerald-700' : 'bg-red-100 text-red-700'}`}>
        {g.gap === 0 ? 'Satisfied' : `Gap: ${g.gap}`}
      </span>
    </div>
  )}}
</div>
</div>
)}

```

```

{roleMode === 'learner' && activeTab === 'recommendations' && (
  <div className="bg-white rounded-2xl p-6 border border-slate-200 shadow-sm space-y-4">
    <h1 className="text-xl font-bold text-slate-900">Learning Recommendations Catalogue</h1>
    <div className="grid grid-cols-1 md:grid-cols-2 gap-4 pt-2">
      {recommendations.map((rec, idx) => (
        <div key={idx} className="border border-slate-200 rounded-xl p-5 bg-slate-50 space-y-3">
          <div className="flex items-center justify-between">
            <span className="text-xs font-bold text-blue-700 bg-blue-100 px-2 py-0.5 rounded">
              Score: {rec.score}/100
            </span>
            <span className="text-xs text-slate-500">{rec.duration}</span>
          </div>
        </div>
      )
    )}
  </div>
)}

```

```

    <h3 className="text-base font-bold text-slate-900">{rec.course_title}</h3>
    <p className="text-xs text-slate-600 bg-white p-2.5 rounded-lg border border-slate-200">
      "{rec.explainability.recommendation_reason}"
    </p>

    <div className="pt-2 flex items-center justify-between">
      <button
        onClick={() => handleOpenExplainability(rec)}
        className="text-xs font-bold text-blue-600 hover:underline"
      >
        Why this course? (Explainability Trail)
      </button>
      <button className="px-3 py-1 bg-slate-900 text-white rounded text-xs font-semibold">
        Enroll
      </button>
    </div>
  </div>
  )})
</div>
})

```

```

{activeTab === 'assessments' && (
  <div className="bg-white rounded-2xl p-6 border border-slate-200 shadow-sm space-y-6">
    <div>
      <h1 className="text-xl font-bold text-slate-900">Assessment Generator & Quiz Module</h1>
      <p className="text-xs text-slate-500">Generate structured MCQs from training material</p>
    </div>

    <div className="border-2 border-dashed border-slate-300 rounded-2xl p-8 text-center bg-slate-50 space-y-3">
      <Upload className="w-10 h-10 text-slate-400 mx-auto" />
    </div>
  </div>
)

```

```
<h3 className="text-sm font-bold text-slate-800">Upload NSSTA Training Material (PDF, DOCX, TXT)</h3>
```

```
<button
  onClick={handleGenerateQuizSample}
  disabled={isGeneratingQuiz}
  className="px-4 py-2 bg-blue-600 hover:bg-blue-700 text-white text-xs font-bold rounded-xl shadow transition"
>
  {isGeneratingQuiz ? 'Extracting & Generating MCQs...' : 'Generate Sample Assessment Quiz'}
</button>
</div>
```

```
{quizQuestions.length > 0 && (
  <div className="space-y-4 pt-4 border-t border-slate-200">
    <h3 className="text-base font-bold text-slate-900">Generated Questions ({quizQuestions.length})</h3>
    <div className="space-y-4">
      {quizQuestions.map((q, idx) => (
        <div key={idx} className="p-4 rounded-xl border border-slate-200 bg-slate-50 space-y-2">
          <span className="text-xs font-bold text-blue-700 uppercase">Question {idx + 1} - {q.competency_name}</span>
          <h4 className="text-sm font-bold text-slate-900">{q.question}</h4>
          <div className="grid grid-cols-2 gap-2 pt-2">
            {q.options.map((opt: string, oldx: number) => (
              <div key={oldx} className={`p-2 rounded text-xs border ${
                oldx === q.correct_answer ? 'bg-emerald-100 border-emerald-300 text-emerald-900 font-bold' : 'bg-white border-slate-200 text-slate-700'
              }`}>
                {String.fromCharCode(65 + oldx)}. {opt}
              </div>
            ))}
          </div>
          <p className="text-xs text-slate-500 pt-2 font-medium italic">Explanation: {q.explanation}</p>
        </div>
      ))}
    </div>
  )}
```

</div>

</div>

)}

</div>

)}

</main>

</div>

{/\* Unique Feature 1: Explainability Trail Modal \*/}

<ExplainabilityModal

  isOpen={isExplainModalOpen}

  onClose={() => setIsExplainModalOpen(false)}

  data={selectedExplainability}

/>

</div>

);

}